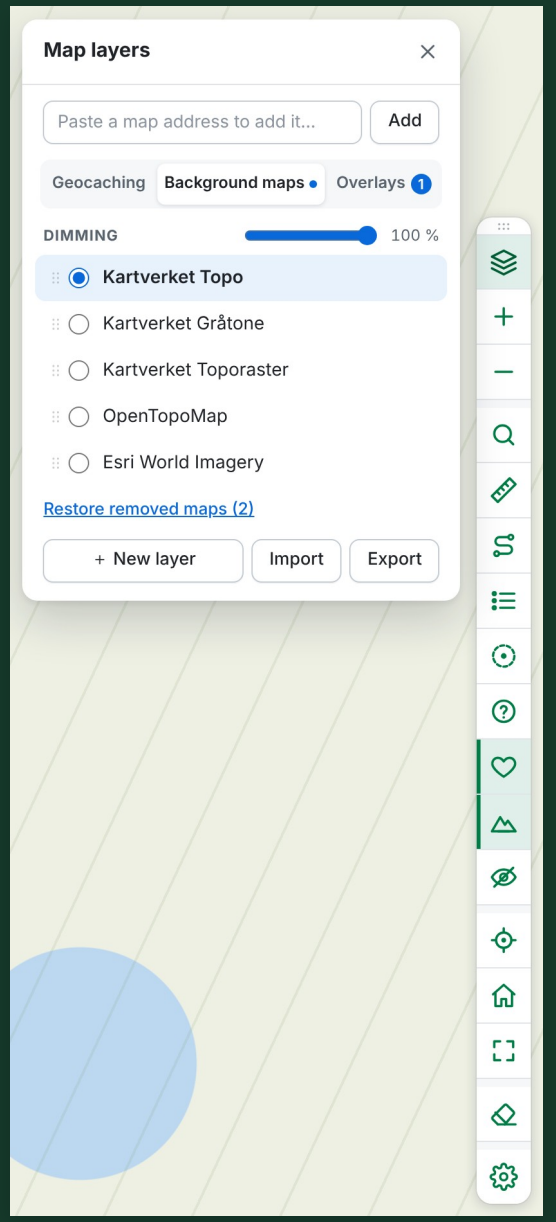
# Geocaching Map – Custom Tile Layers

Your own maps and handy tools right in the geocaching.com maps

Userscript · version 1.53.0 · English and Norwegian



# Everything you need on the map, in one toolbar

### Any map

Kartverket, OpenStreetMap, aerial photos – or your own XYZ, WMTS, WMS and ArcGIS services

### Place search

Place names and addresses from Kartverket, places abroad and coordinates – in one list

### Coordinates and projecting

DMM, DD, DMS and UTM (32, 33, 35), projecting, and measuring distance and bearing

### 161 m circles

Around traditional and solved caches and visible physical stages – with clickable marks

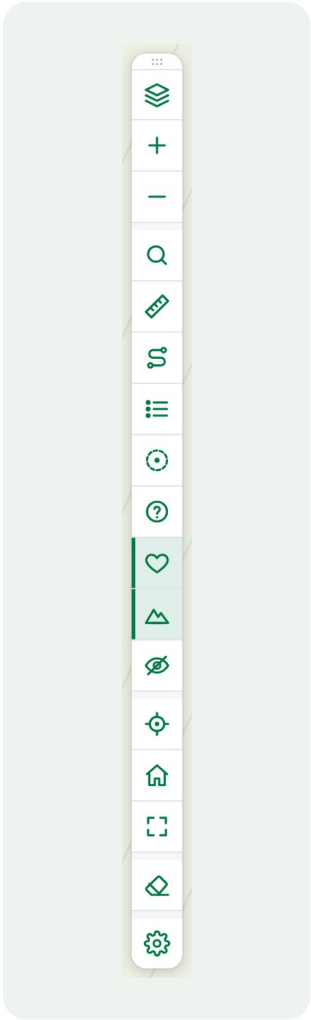
### Home, location, Street View

Go home or to your location, and open Street View from the right-click menu

### New: routes for your GPS

Pick caches and pins from a list or on the map, preview and download as a GPX route

# One toolbar instead of many buttons



### Map layers
Choose background map and overlays

### Find a place
Shortcut: press / on the map

### Measure
Distance and bearing between points

### Route
Build a route and download it as GPX

### Geocaches on the map
A list that follows the map (/map)

### 161 m circles
Where something is placed – from zoom 15

### Marks
Solved caches, also without circles

### Favourite points
❤️ and the count above the caches – on/off

### D/T
Difficulty/terrain below the caches – on/off

### Hide geocaches
A clean map – click again to show them

### My location
With an accuracy circle

### Home
Right-click to change the address

### Fullscreen
The map fills the whole screen

### Clear drawings
Marks, measurements and circles

### Settings
Position, language and the site's buttons

# Choose maps and overlays

### Three tabs

Geocaching.com's own maps, background maps and overlays – each on its own, so the list stays short

### Background and overlays

Kartverket, OpenTopoMap, OpenStreetMap and Esri – plus your own. As many overlays as you like

### Fading towards white

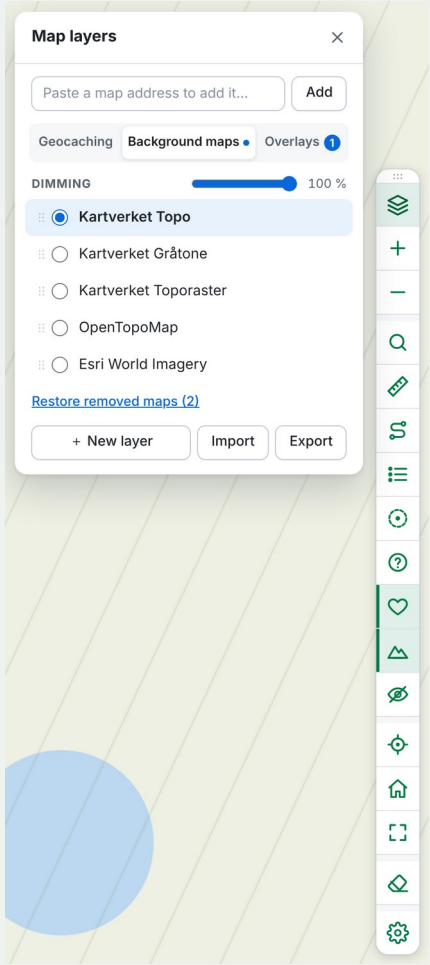
The background can be faded by up to 50 %, so the caches stand out

### Your order

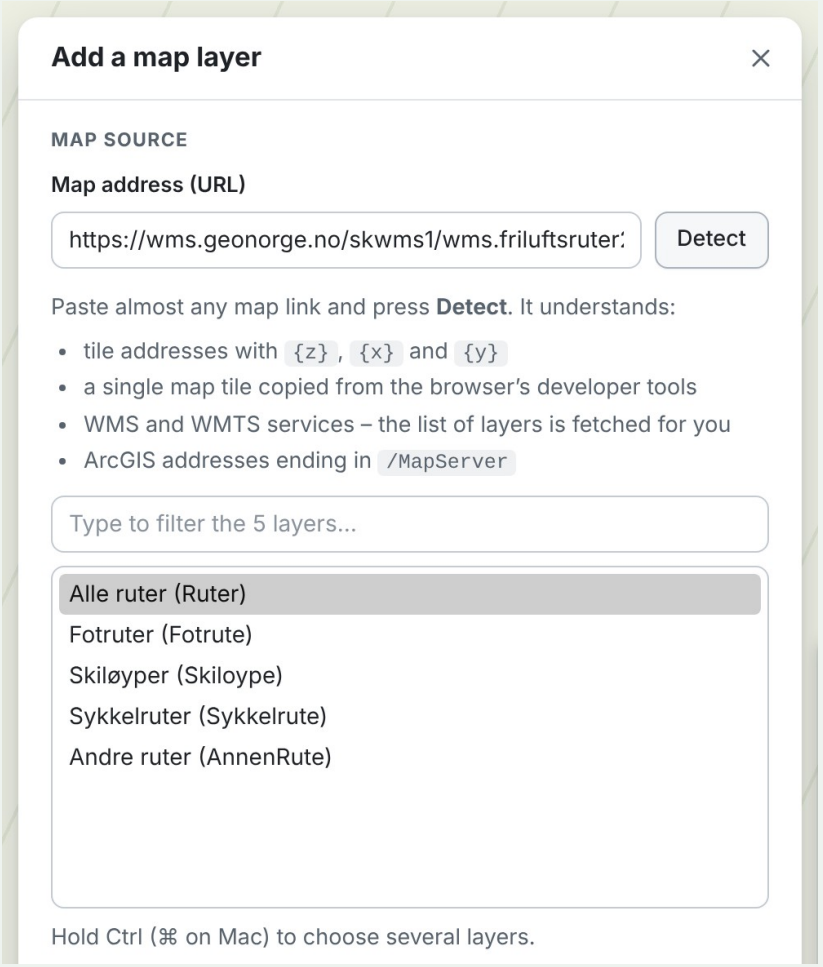
Drag the maps up and down. The overlay at the top of the list is drawn on top

### Tidy the list

Remove maps you don't use – also geocaching.com's own – and bring them back whenever you like



# Add your own maps




Paste almost any map link and press Detect:

**XYZ / TMS**
Tile addresses with {z} {x} {y}

**WMTS**
The list of layers is fetched automatically

**WMS**
Pick one or more layers

**ArcGIS**
Addresses ending in /MapServer

- Preview on the map before you save
- Import and export your layers as a file
- A single map tile from the developer tools is enough

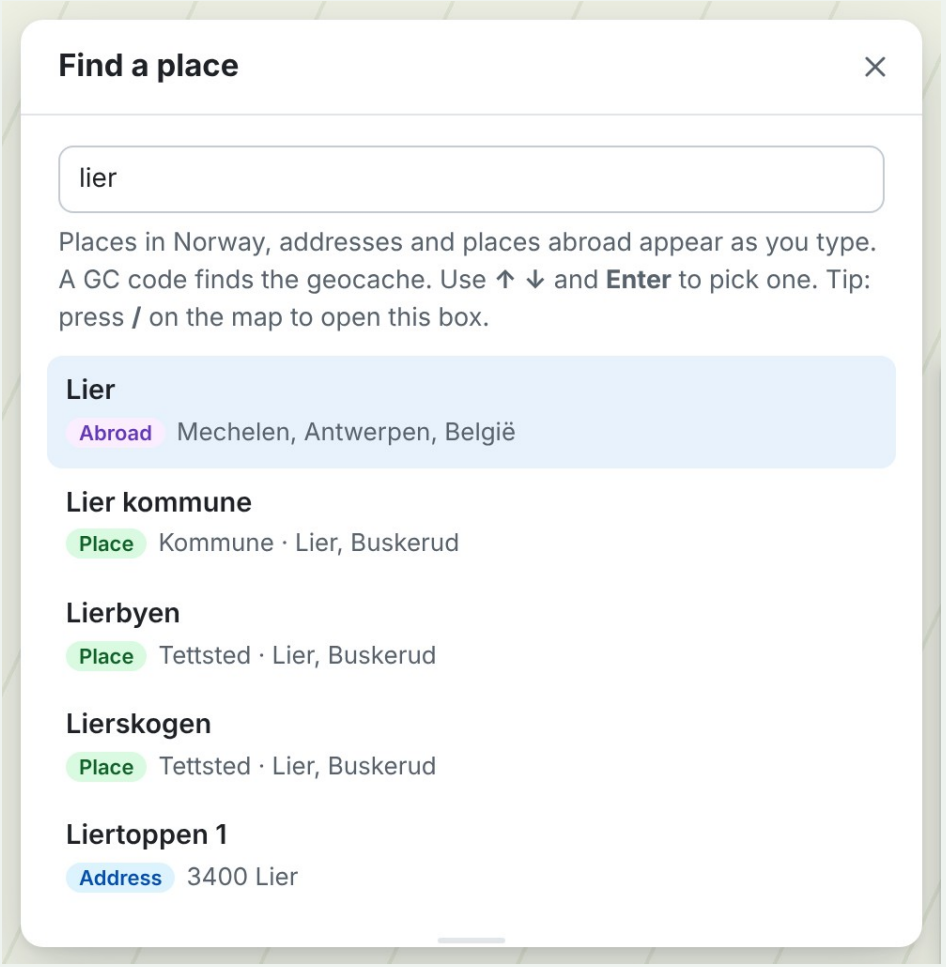
# Find places, addresses and coordinates

### One combined list
Hits appear as you type, best match first – whatever the source

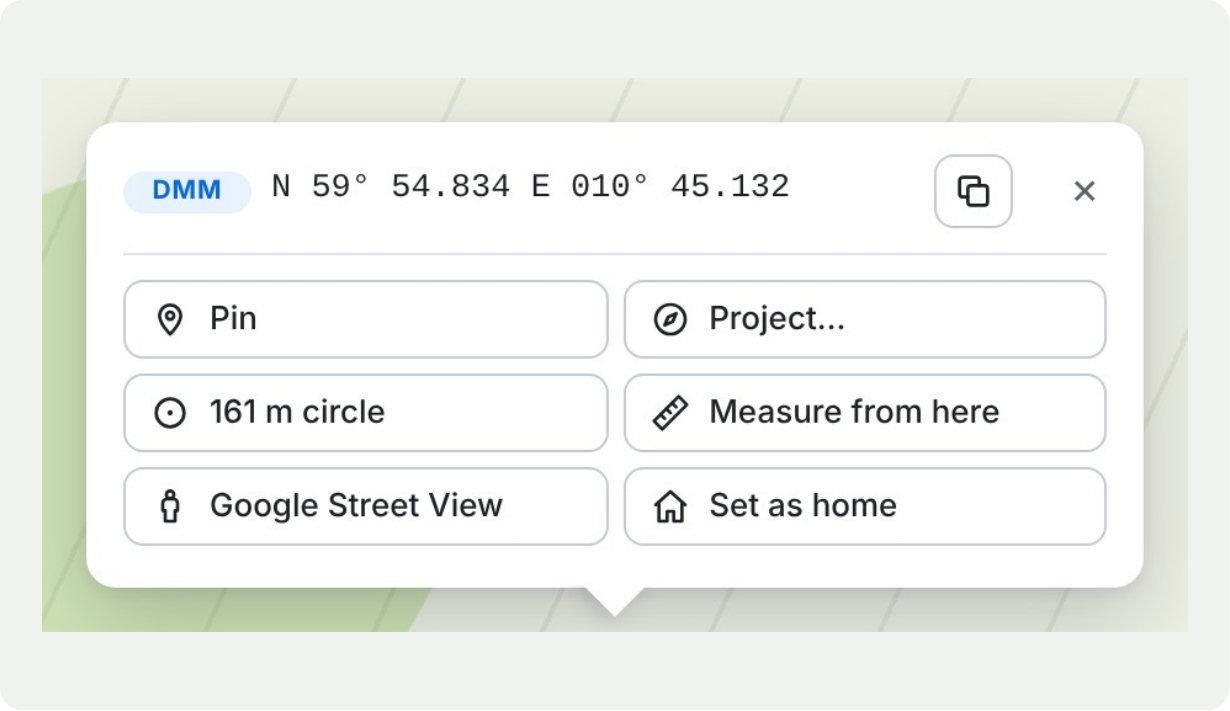
### Places, addresses and abroad
Kartverket's place names and addresses, and places abroad via Photon (OpenStreetMap)

### Coordinates
Type DMM, DD, DMS or UTM – the script recognises them

### Home address
The same search box is used to save your home address



# Coordinates in every format




| Format | Example (Oslo) |
|---|---|
| DMM | `N 59° 54.834 E 010° 45.132` |
| DD | `59.913900, 10.752200` |
| DMS | `N 59° 54' 50.0" E 10° 45' 07.9"` |
| UTM | `32V 597980E 6643119N` |
| UTM 33 | `33V 262560E 6649444N` |

Right-click the map: one coordinate line that opens in DMM. Click the line to switch format, and copy with one click. The Google Street View button opens Street View at the spot. The zoom level is at the bottom – type a number to zoom.

UTM 32, 33 and 35 with a fixed zone, as Norwegian maps use. Accurate to about a millimetre (Krüger series).

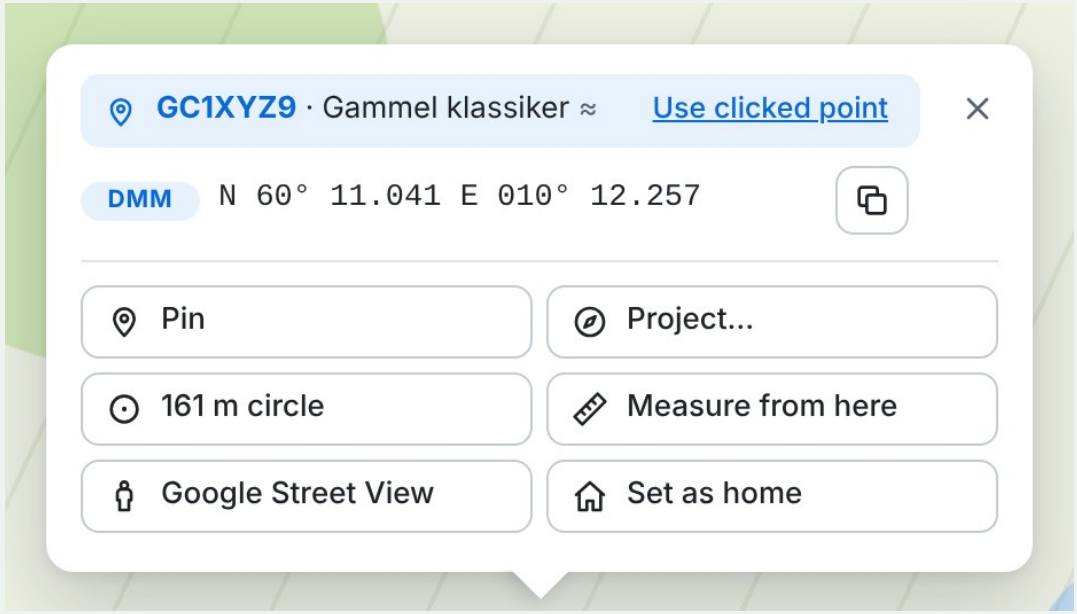
# Right-click a cache – straight to its coordinates

### Snaps to the cache
Right-click on or right next to a geocache, and the popup uses the cache's coordinates

### GC code, name and ❤️
Favourite points are shown; click the GC code to open the cache page

### All tools come along
Pin, projecting, measuring and the 161 m circle start from the cache

### Use the clicked spot
One click switches to the exact spot you clicked




Exact coordinates from geocaching.com's cache search – also on /map. Only before the search has answered is the position worked out from the icon (≈).

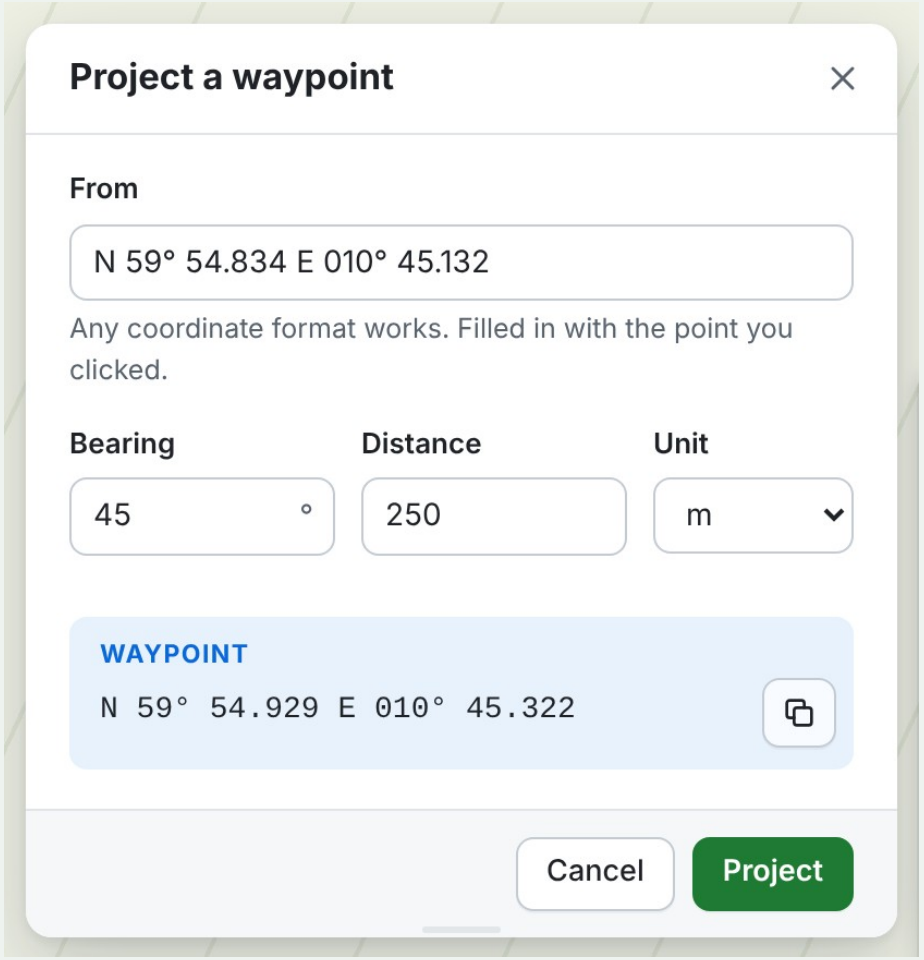
# Project waypoints and measure distances

### Project from a point
The start point is filled in for you. Type bearing and distance in m, km or ft

### See the result at once
The waypoint shows as you type, with a dashed line on the map and a copy button

### Measure distance and bearing
Click points on the map – total length and the last leg are shown

### Good enough for geocaching
The error is well below a metre at ordinary distances



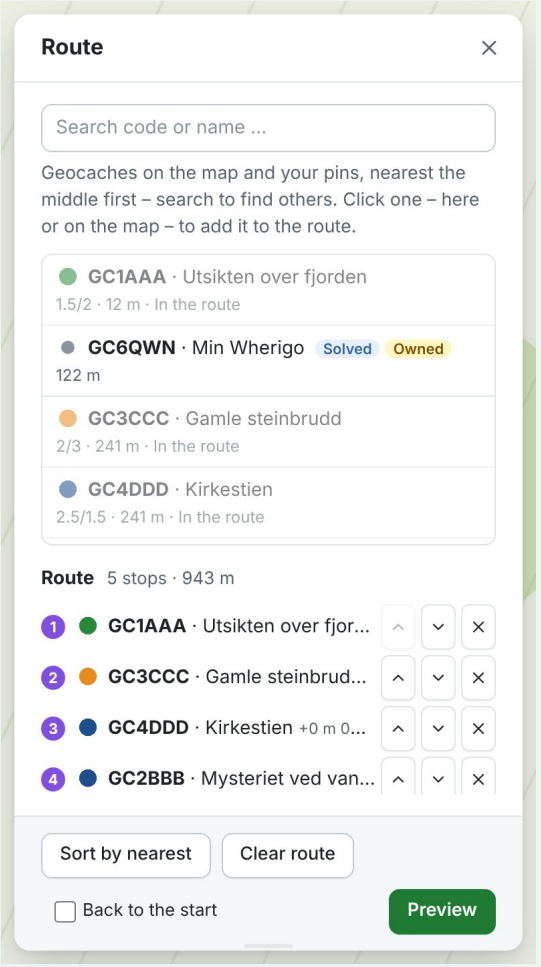
# New: build a route and take it with you on your GPS

### Searchable list
Nearby geocaches and pins – and any GC code worldwide

### Pick on the map
Click caches and pins while the route window is open, or use the right-click popup

### Order and round trip
Move stops, sort by nearest, see every leg – and end at the start

### Preview and download
GPX with Garmin symbols – optionally with full cache data (paperless caches)



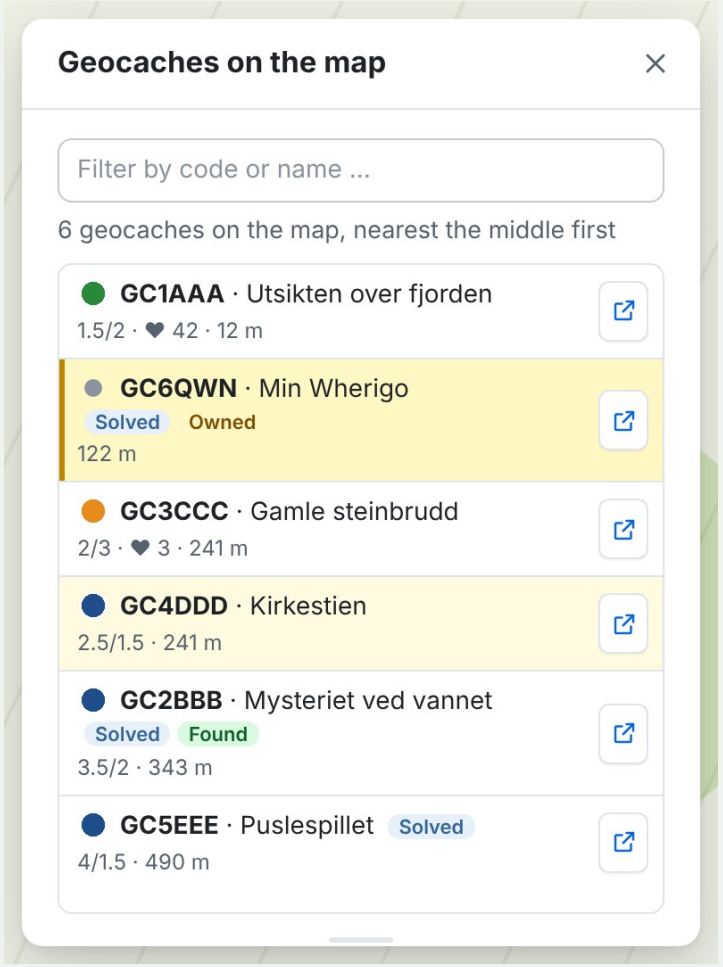
# New: the geocaches on the map in a list

### Follows the map
The geocaches on the map you see, nearest the middle first – with Solved, Found, Owned, D/T and ❤️

### Point and find
Point at a cache on the map and it is highlighted in the list. Filter by code or name

### Zoom with a glow
Click to zoom all the way in – the cache glows on the map. Click again to go back

### The cache page
The button on the line opens the cache page in a new tab



# 161 m circles where something is placed

### Traditional caches
At their exact posted coordinates

### Solved caches
At your corrected coordinates – also when the icon has moved back after a find

### Physical stages and finals
Visible stages of multis, mysteries, Wherigos and letterboxes, read from the cache page

### Single circle
Right-click anywhere and choose 161 m circle



Where no cache icon shows the spot, a clickable mark sits in the middle of the circle. Hovering over a solved cache shows “Final location”; right-click gives the exact coordinates.

# 161 m circles, cache pages and home



## 161 m circles

- Where something is placed – see the previous slide
- Shown from zoom level 15
- Follow the map and never block clicks
- Single circle: right-click



## Cache pages

- The small map shows your chosen map
- Same fading and overlays
- Keeps the site's view
- Never changes your choice



## Home

- One click takes you home
- The first time you look up the address
- Stored only in your browser

# How the map layers are stacked

### ABOVE THE MAP TILES

| | | |
|---|---|---|
| **Controls** | `controls · 1000` | the toolbar, the layer panel, the coordinate display and the credits |
| **Popups** | `popupPane · 700` | the right-click popup and other map popups |
| **Markers** | `markerPane · 600` | the cache icons on /play/map, pins, labels and your location |
| **Drawings** | `overlayPane · 400` | 161 m circles, measuring lines, projection lines and the accuracy circle |

### THE MAP TILES (TILEPANE · 200) – SORTED BY Z-INDEX

| | | |
|---|---|---|
| **The site's other tiles** | `z = bz + 10` | the cache icons on /map – always lifted above your layers |
| **Preview** | `z = bz + 3` | the layer you are trying in the layer editor |
| **Your overlays** | `z = bz + 2` | hiking routes, property boundaries, hillshade – their own opacity slider |
| **Your background map** | `z = bz + 1` | fully opaque – faded towards white with a colour filter, at most 50 % |
| **The geocaching.com map** | `z = bz` | the site's background – covered by your map, faded when it is chosen itself |

bz = the z-index of the site's background. On /play/map the site's background is a vector map (WebGL). On /map an invisible cache grid (UTFGrid) is loaded as well, used for the 161 m circles and for snapping the right-click popup to caches.

# Under the hood

### Never removes the site's layers
The script slots its own layers in between, sorted by z-index

### Finds the map by itself
Catches Leaflet as the page loads, also on /play/map and the cache pages

### No seams between the tiles
Tiles are drawn 1 pixel larger, also at 110 and 125 % display scaling

### Leaves the form alone
No button can submit the page's ASP.NET form, so the page never reloads

### Fading without show-through
A colour filter bleaches the map towards white – the site's map never shows through

### Built-in troubleshooting
gcctReport(), gcctCacheDebug(), gcctStageTest() and gcctSnapTest() in the console give short reports without personal data

# Get started in three minutes

1

## Install a userscript manager

Tampermonkey or Violentmonkey in Chrome, Edge, Firefox or Safari

2

## Add the script

Open geocaching-custom-tiles.user.js and save it in the manager

3

## Open a map

geocaching.com/map or /play/map – the toolbar appears on the right

## Privacy

The script collects and sends no personal data and has no server of its own. It only contacts the map servers you switch on, Kartverket and Photon when you search, and geocaching.com's own cache search and cache pages when the 161 m circles are on. Your location is only requested when you ask for it. All settings are stored in the browser.

# Better maps for better trips

User guides in English and Norwegian are included. Use gcctReport() in the browser console if something doesn't work.

